

# Algebraic Geometry – Exercise Sheet 3

Week 3: Schemes, morphisms, gluing and first global properties  
Monday and Thursday afternoon sessions; Lectures 11–15

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**Difficulty.** ★ routine; ★★ synthesis; ★★★ challenge or extension.

**Use.** The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

**Hints.** Read them progressively and stop as soon as one is enough to restart the problem.

## Monday afternoon — material available: Lectures 1–11

### Exercise 1. Global functions as morphisms to the affine line

★ **CORE**

Let  $X$  be a scheme.

- (i) Explain why a global section  $s \in \Gamma(X, \mathcal{O}_X)$  determines a morphism

$$f_s : X \longrightarrow \mathbb{A}_{\mathbb{Z}}^1.$$

- (ii) Show that

$$\mathrm{Hom}(X, \mathbb{A}_{\mathbb{Z}}^1) \simeq \Gamma(X, \mathcal{O}_X).$$

- (iii) If  $X = \mathrm{Spec}(A)$ , describe the morphism associated with  $a \in A$ .

**Hint 1 (conceptual).** Use the affine-target adjunction for  $\mathbb{A}_{\mathbb{Z}}^1 = \mathrm{Spec}(\mathbb{Z}[t])$ .

**Hint 2 (technical).** A unital ring homomorphism  $\mathbb{Z}[t] \rightarrow \Gamma(X, \mathcal{O}_X)$  is uniquely determined by the image of  $t$ .

### Exercise 2. The reduced point and the double point

★ **CORE**

Inside  $\mathbb{A}_{\mathbb{k}}^1 = \mathrm{Spec}(\mathbb{k}[t])$ , consider

$$P = \mathrm{Spec}(\mathbb{k}[t]/(t)), \quad D = \mathrm{Spec}(\mathbb{k}[t]/(t^2)).$$

- (i) Show that the two closed immersions have the same image as subsets of  $\mathbb{A}_{\mathbb{k}}^1$ .  
(ii) Show that  $P$  and  $D$  are not isomorphic as schemes.  
(iii) Compute the maps on local rings at their unique points.

**Hint 1 (conceptual).** The two images as subsets are  $V(t)$  and  $V(t^2)$ . Compare their radicals.

**Hint 2 (technical).** Each source has only one point, so its local ring there is its whole coordinate ring. Reducedness already distinguishes the two schemes.

### Exercise 3. A morphism between affine schemes

★ **CORE**

Let

$$\varphi : \mathbb{k}[u, v] \longrightarrow \mathbb{k}[t], \quad u \longmapsto t, \quad v \longmapsto t^2.$$

- (i) Describe the induced morphism  $f : \mathbb{A}_{\mathbb{k}}^1 \rightarrow \mathbb{A}_{\mathbb{k}}^2$ .  
(ii) Determine  $\ker(\varphi)$  and factor  $f$  through a closed subscheme of  $\mathbb{A}_{\mathbb{k}}^2$ .  
(iii) Show that the resulting morphism onto that closed subscheme is an isomorphism.

**Hint 1 (technical).** The relation  $v - u^2$  is in the kernel. Divide an arbitrary  $F(u, v)$  by the monic polynomial  $v - u^2$ , viewing it as a polynomial in  $v$ .

**Hint 2 (technical).** After passing to  $\mathbb{k}[u, v]/(v - u^2)$ , write down an inverse to the induced map by specifying the image of  $t$ .

### Exercise 4. An arithmetic spectrum

★★ **CHOICE**

Let  $A = \mathbb{Z}[s]/(s^2 - 3)$ . Describe the inverse image in  $\mathrm{Spec}(A)$  of the points  $(2)$ ,  $(3)$ , and  $(p)$  for  $p \neq 2, 3$ , according to the factorisation of  $s^2 - 3$  modulo  $p$ .

**Hint 1 (technical).** The inverse image of  $(p)$  is the spectrum of  $\mathbb{F}_p[s]/(s^2 - 3)$ . Factor the polynomial in the three cases  $p = 2$ ,  $p = 3$ , and  $p \neq 2, 3$ .

## Thursday afternoon — material available: Lectures 1–14

### Exercise 5. Dual numbers and tangent vectors

★★ CORE

Let  $D = \text{Spec}(\mathbb{k}[\varepsilon]/(\varepsilon^2))$  and let  $p = (a, b) \in \mathbb{A}_{\mathbb{k}}^2(\mathbb{k})$ .

(i) Show that morphisms  $D \rightarrow \mathbb{A}_{\mathbb{k}}^2$  whose closed point maps to  $p$  are uniquely of the form

$$x \mapsto a + u\varepsilon, \quad y \mapsto b + v\varepsilon.$$

(ii) Let  $C = \mathbb{W}(f) \subset \mathbb{A}_{\mathbb{k}}^2$ . Show that such a morphism factors through  $C$  precisely when

$$f(a, b) = 0, \quad f_x(a, b)u + f_y(a, b)v = 0.$$

(iii) Apply this to the parabola  $y = x^2$ .

**Suggested strategy.** Translate the morphism into a homomorphism of coordinate rings, then expand  $f$  after substituting  $a + u\varepsilon$  and  $b + v\varepsilon$  and discard every term containing  $\varepsilon^2$ .

**Hint 1 (technical).** Every element of  $\mathbb{k}[\varepsilon]/(\varepsilon^2)$  is uniquely  $c + d\varepsilon$ . Mapping the closed point to  $(a, b)$  fixes the constant terms of the images of  $x$  and  $y$ .

**Hint 2 (technical).** Prove the first-order formula

$$f(a + u\varepsilon, b + v\varepsilon) = f(a, b) + (f_x(a, b)u + f_y(a, b)v)\varepsilon.$$

### Exercise 6. The line with two origins

★★ CORE

Glue two copies

$$X_1 = \text{Spec}(\mathbb{k}[t_1]), \quad X_2 = \text{Spec}(\mathbb{k}[t_2])$$

along  $D(t_1) \simeq D(t_2)$  by  $t_1 \mapsto t_2$ . Let  $X$  be the resulting scheme.

(i) Describe the underlying topological space.

(ii) Show that  $\Gamma(X, \mathcal{O}_X) \simeq \mathbb{k}[t]$ .

(iii) Show that the canonical map  $X \rightarrow \text{Spec } \Gamma(X, \mathcal{O}_X)$  identifies the two origins.

(iv) Deduce that  $X$  is not affine.

**Suggested strategy.** Describe the gluing topologically first. Then compute global functions as compatible pairs on the two affine charts. Finally compare the scheme with the spectrum of that global ring.

**Hint 1 (technical).** A global function is a pair  $(f_1, f_2) \in \mathbb{k}[t_1] \times \mathbb{k}[t_2]$  whose images agree in  $\mathbb{k}[t, t^{-1}]$ . Use the injectivity of  $\mathbb{k}[t] \rightarrow \mathbb{k}[t, t^{-1}]$ .

**Hint 2 (conceptual).** The canonical map to  $\text{Spec } \Gamma(X, \mathcal{O}_X)$  restricts to the ordinary coordinate map on both charts. Track the two origins under this map.

### Exercise 7. Global functions on the projective line

★★ CHOICE

Construct  $\mathbb{P}_{\mathbb{k}}^1$  by gluing  $U_0 = \text{Spec}(\mathbb{k}[t])$  and  $U_1 = \text{Spec}(\mathbb{k}[s])$  along  $s = t^{-1}$ . Prove that

$$\Gamma(\mathbb{P}_{\mathbb{k}}^1, \mathcal{O}_{\mathbb{P}^1}) = \mathbb{k}.$$

**Hint 1 (technical).** A global function is a pair  $f(t) \in \mathbb{k}[t]$ ,  $g(s) \in \mathbb{k}[s]$  satisfying  $f(t) = g(t^{-1})$  in  $\mathbb{k}[t, t^{-1}]$ . Compare the possible positive and negative powers of  $t$ .

### Exercise 8. The projective conic from its standard opens

★★ CORE

Let

$$S = \mathbb{k}[x_0, x_1, x_2]/(x_0x_2 - x_1^2), \quad C = \text{Proj}(S).$$

(i) Compute  $(S_{x_0})_0$  and  $(S_{x_2})_0$ .

(ii) Identify  $D_+(x_0)$  and  $D_+(x_2)$  with affine lines.

(iii) Describe their overlap and recover the gluing rule for  $\mathbb{P}^1$ .

**Suggested strategy.** Compute the degree-zero parts of the two homogeneous localisations, then express the transition function on the overlap.

**Hint 1 (technical).** On  $D_+(x_0)$  use  $t = x_1/x_0$  and the relation  $x_2/x_0 = t^2$ . Do the symmetric computation on  $D_+(x_2)$ .

**Hint 2 (technical).** On the overlap, both  $x_0$  and  $x_2$  are non-zero. Compare the two affine parameters using  $x_0x_2 = x_1^2$ .

### Exercise 9. Connectedness and idempotents

★★ **CORE**

- (i) Prove that  $\text{Spec}(A)$  is disconnected if and only if  $A$  has a non-trivial idempotent.
- (ii) Apply this to  $A = \mathbb{k}[t]/(t^2 - t)$ .
- (iii) Show that  $\text{Spec}(\mathbb{k}[x, y]/(xy))$  is connected although reducible.

**Hint 1 (conceptual).** A decomposition of  $\text{Spec}(A)$  into two disjoint open-and-closed subsets should correspond to a function taking the values 0 and 1 on the two pieces.

**Hint 2 (technical).** Starting from an idempotent  $e$ , use the two principal opens  $D(e)$  and  $D(1 - e)$ . Conversely, translate a clopen decomposition into a product decomposition of  $A$ .

**Hint 3 (technical).** For  $\mathbb{k}[x, y]/(xy)$ , an idempotent restricts to idempotents on both axes. Compare their values at the common origin.

### Exercise 10. Reduction and its universal property

★★ **CORE**

Let

$$A = \mathbb{k}[x, y]/(x^2, xy), \quad X = \text{Spec}(A).$$

- (i) Compute the nilradical and  $X_{\text{red}}$ .
- (ii) Let  $R$  be a reduced  $\mathbb{k}$ -algebra. Show that every homomorphism  $A \rightarrow R$  factors uniquely through  $A_{\text{red}}$ .
- (iii) Translate this into a statement about morphisms of schemes.

**Hint 1 (conceptual).** Find the nilradical first. If  $R$  is reduced, the image in  $R$  of every nilpotent element of  $A$  must vanish.

**Hint 2 (technical).** Use this to show that the kernel of any homomorphism  $A \rightarrow R$  contains the nilradical. The desired factorisation is then the universal property of a quotient ring.

**Hint 3 (conceptual).** Remember to reverse arrows when translating the factorisation into a statement about schemes.

### Exercise 11. A weighted projective chart

★★★ **OPTIONAL**

Let  $R = \mathbb{k}[x, y, z]$  with  $\deg x = \deg y = 1$  and  $\deg z = 2$ , and set  $X = \text{Proj}(R)$ . Show that  $D_+(x)$  and  $D_+(y)$  are affine planes and that

$$D_+(z) \simeq \text{Spec}(\mathbb{k}[a, b, c]/(ac - b^2)).$$

Identify the singular point of this chart.

**Suggested strategy.** For each standard open  $D_+(f)$ , compute the degree-zero subring  $(R_f)_0$ . The chart at  $z$  is the only one where the grading produces a relation.

**Hint 1 (technical).** On  $D_+(z)$  consider the degree-zero elements

$$a = x^2/z, \quad b = xy/z, \quad c = y^2/z.$$

They satisfy one evident relation.

**Hint 2 (technical).** To locate the singular point of  $ac - b^2 = 0$ , use the partial derivatives with respect to  $a, b, c$ .

## After Friday's lecture or preparation for Week 4

### Exercise 12. Components and generic points

★★ **CHOICE**

Let

$$I = (xy, xz) \subset \mathbb{k}[x, y, z], \quad X = \operatorname{Spec}(\mathbb{k}[x, y, z]/I).$$

Show that  $I = (x) \cap (y, z)$ . Determine the irreducible components, their generic points and dimensions, and decide whether  $X$  is connected or integral.

**Hint 1 (conceptual).** For an affine scheme, irreducible components correspond to minimal prime ideals, and their generic points are those minimal primes viewed as points of the spectrum.

**Hint 2 (technical).** To prove  $(x) \cap (y, z) = (xy, xz)$ , write an element of the intersection as  $xg$  and reduce modulo  $(y, z)$ .

**Hint 3 (conceptual).** Use the definition of irreducibility and inspect whether the two components meet. Reducedness comes from the expression of  $I$  as an intersection of prime ideals.

**Exercise 13. The function field of the cusp**

★★ **CHOICE**

Let

$$C = \operatorname{Spec}(\mathbb{k}[x, y]/(y^2 - x^3)).$$

Show that  $C$  is integral and that  $K(C) \simeq \mathbb{k}(t)$ .

**Hint 1 (technical).** Use the map  $x \mapsto t^2, y \mapsto t^3$ . Every class in the coordinate ring has a representative  $A(x) + yB(x)$ ; parity of exponents proves injectivity.

**Hint 2 (technical).** The fraction field is contained in  $\mathbb{k}(t)$ . Recover  $t$  as a quotient of the images of  $x$  and  $y$ .